The NOR function truth table is shown below:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

A NOR B = NOT (A OR B)

2.9 Basic Logic Gates:

A gate is a basic electronic circuit which operates on one or more signals to produce an output signal. There are three fundamental gates namely AND, OR and NOT. The other logic gates like NAND, NOR, XOR and XNOR are derived gates which are derived from the fundamental gates. NAND and NOR gates are called Universal gates, because the fundamental logic gates can be realized through them.

2.9.1 AND Gate

The AND gate can have two or more input signals and produce an output signal.

The output is "true" only when both inputs are "true", otherwise, the output is "false". In other words the output will be 1 if and only if both inputs are 1; otherwise the output is 0. The output of the AND gate is represented by avariable say C, where A and B are two boolean variables. In boolean algebra, a variable can take either of the values '0' or '1'. The logical symobl of the AND gate is

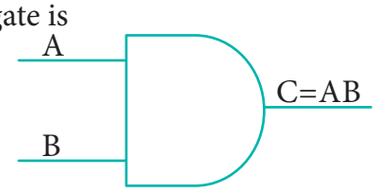


Fig. 2.4 Logic symbol of AND Gate

One way to symbolize the action of an AND gate is by writing the boolean function.

C = A AND B

In boolean algebra the multiplication sign stands for the AND operation. Therefore, the output of the AND gate is

C = A . B or

simply C = AB

Read this as "C equals A AND B". Since there are two input variables here, the truth table has four entries, because there are four possible inputs : 00, 01, 10 and 11.

For instance if both inputs are 0,

C = A . B

= 0 . 0

= 0

The truth table for AND Gate is

Input		Output
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

Table 2.7 Truth Table for AND Gate

2.9.2 OR Gate

The OR gate gets its name from its behaviour like the logical inclusive "OR". The output is "true" if either or both of the inputs are "true". If both inputs are "false" then the output is "false". In otherwords the output will be 1 if and only if one or both inputs are 1; otherwise, the output is 0. The logical symbol of the OR gate is

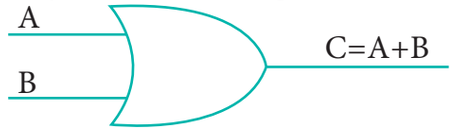


Fig. 2.5 Logic symbol of OR Gate

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